ELECTROMAGNETIC WAVES [18EC55]

- TIE ECE Review Team

Module - 01

01. Coulomb's law - vector form & statement.
02. Electric field intensity due to infinite line charge & surface or sheet of charge.
03. Problems on Coulomb's law.
04. Problems on Electric field intensity.

Module - 02

01. Gauss law - Statement & proof.
02. Applications of Gauss law - coaxial cable
    - spherical shell of charge
03. Relation b/n D & E.
04. Divergence theorem - statement & proof.
05. Relation b/n Electric field intensity & electric potential [$E = -\nabla V$], Energy expended in moving a charge, Equation of continuity, current density.
06. Energy Density $= \frac{1}{2}\varepsilon_0 E^2$ [proof]
07. Problems on Current, Energy and workdone.
08. Numericals on $E = -\nabla V$ in three co-ordinates.

Module - 03

01. Statement and Proof of the following:
    (a) Poisson's and Laplace equation.
    (b) Uniqueness theorem
    (c) Biot-Savart law
    (d) Ampere's law
    (e) Stoke's theorem
02. Scalar and Vector magnetic potentials.
03. Problems on curl in three co-ordinates.

TIE-ECE REVIEW TEAM

Scanned with CamScanner

04. Applications - Parallel plate capacitor
- Co-axial cable / cylinder
- Concentric spheres.

05. Problems on verification of Laplace equations.

Module - 04

01. Lorentz force equation - Statement & proof.

02. Faraday's law - point form
- integral form

03. Magnetic Boundary conditions - tangential component.
- normal component.

04. Problems on Faraday's law, lorentz force equation.

05. Force on a differential current element & problems.

06. Magnetization and Permeability [all definitions].

Module - 05

01. Maxwell's equations in static field and time varying field.

02. Maxwell's equations in point & integral form for time varying fields & static field.

03. Wave equations in terms of electric & magnetic fields, for conducting medium, for lossy dielectric or imperfect dielectric.

04. Wave propagation in good conductors, dielectrics and freespace.

05. Boundary conditions, Relation b/n E & H for uniform plane waves.

06. Skin effect, depth of penetration, displacement current density, poynting vector & power flow of problems.

07. $\nabla \times H = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

TIE-ECE REVIEW TEAM

Scanned with CamScanner

08. Problems on uniform plane waves for finding velocity, wavelength, intrinsic impedance, rms value of electric field for all the medium.

TIE-ECE REVIEW TEAM

Scanned with CamScanner